page 1 of 6

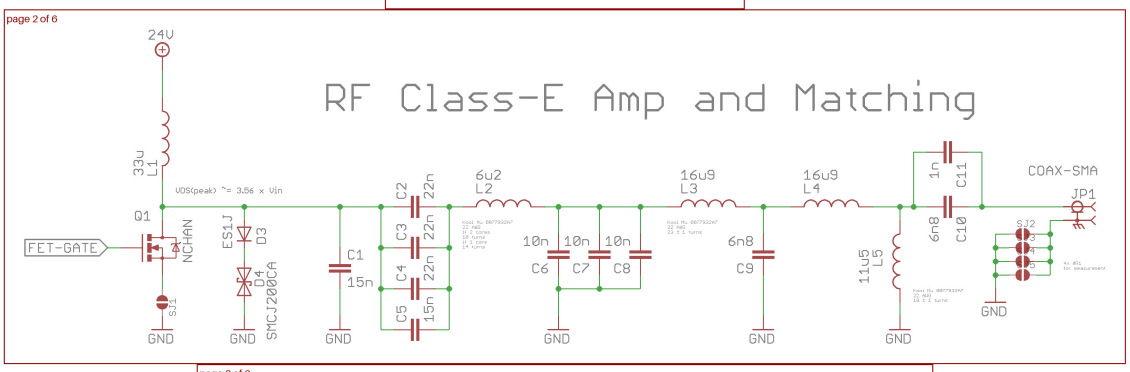
HOT WAND LITE

20260906A

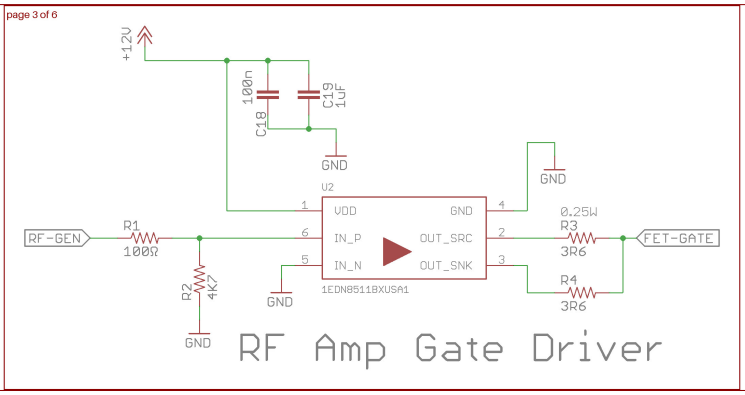
PCB

GND

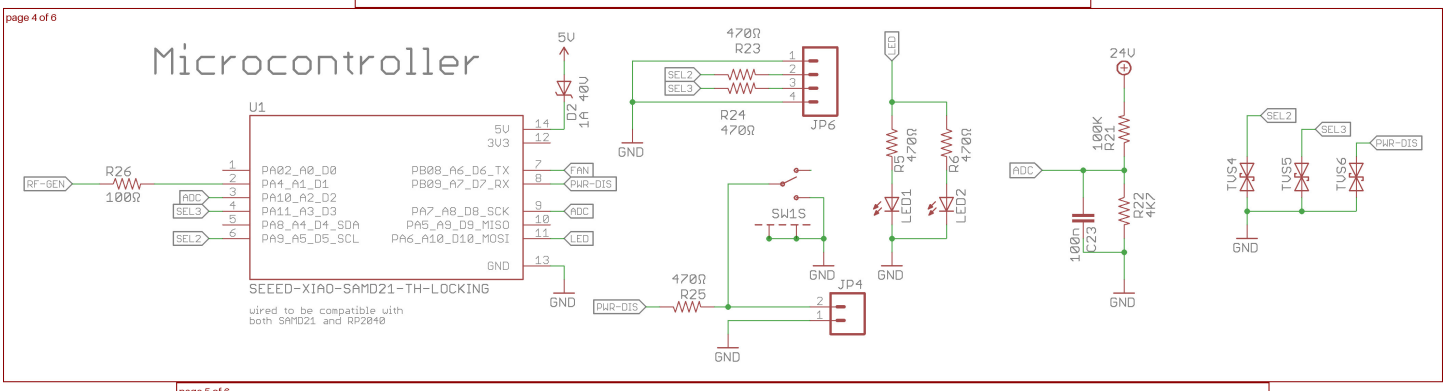
page 2 of 6



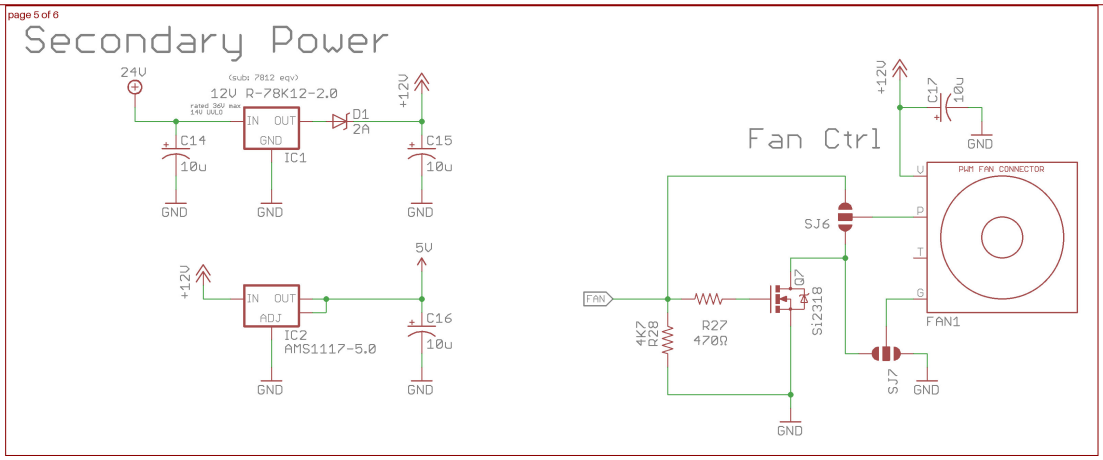

page 3 of 6

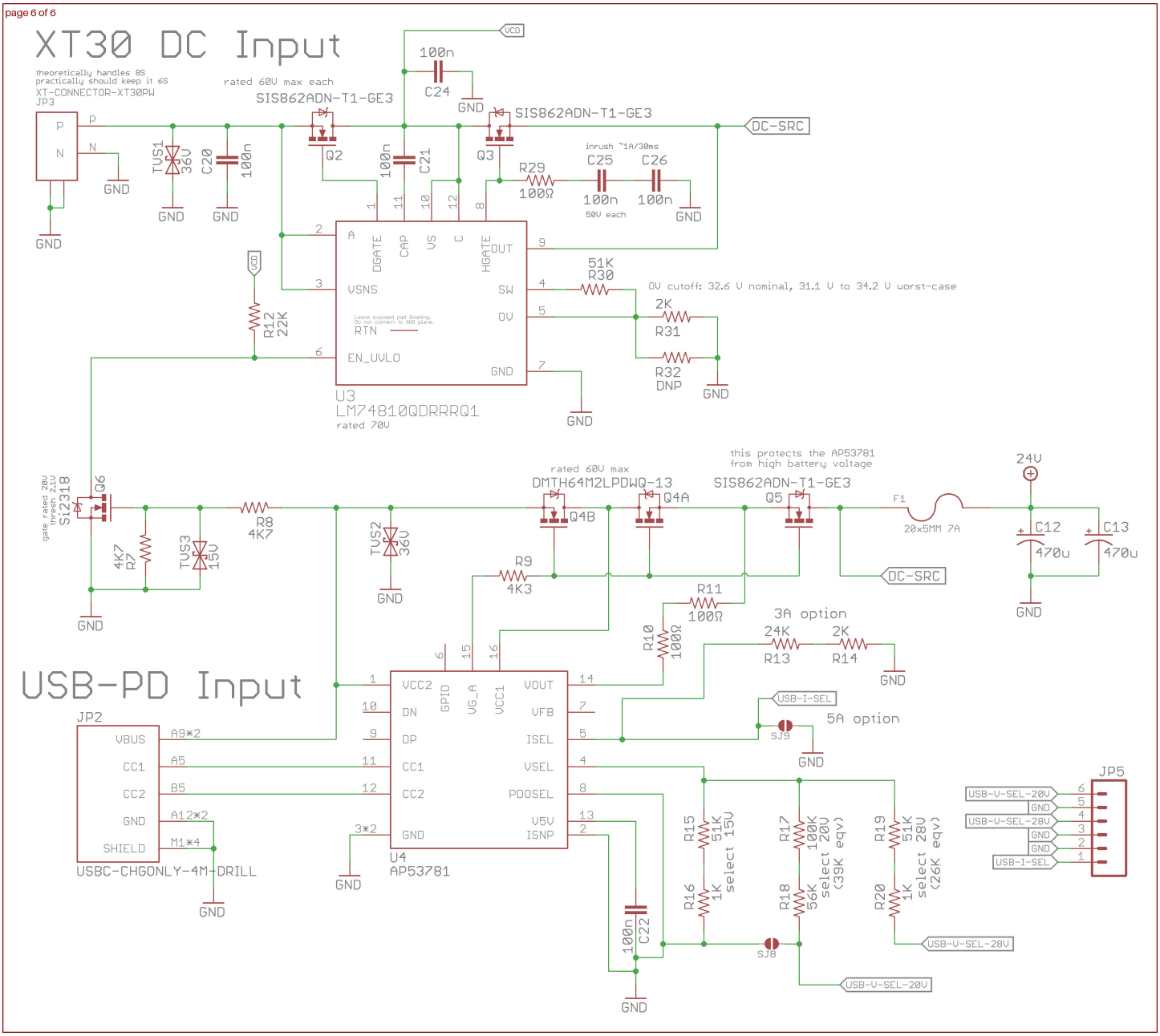



page 4 of 6




page 5 of 6




XT30 DC Input
theoretically handles 8S
practically should keep it 6S
XT-CONNECTOR-XT30PW
JP3
P
N
GND
TVS1
36V
C20
100n
rated 60V max each
SIS862ADN-T1-GE3
Q2
C21
100n
Q3
SIS862ADN-T1-GE3
C24
100n
VCC
DC-SRC
R29
100Ω
inrush ~1A/30ms
C25
100n
C26
100n
50V each
U3
LM74810QDRRRQ1
rated 70V
DGATE
CAP
VS
C
HGATE
OUT
VSNS
SW
OV
RTN
EN_UVLO
GND
R12
22K
51K
R30
2K
R31
R32
DNP
OV cutoff: 32.6 V nominal, 31.1 V to 34.2 V worst-case
Leave exposed pad floating.
Do not connect to GND plane.
USB-PD Input
JP2
VBUS
CC1
CC2
GND
SHIELD
A9*2
A5
B5
A12*2
M1*4
USBC-CHGONLY-4M-DRILL
U4
AP53781
VCC2
DN
DP
CC1
CC2
GND
GPIO
VG_A
VCC1
VOUT
VFB
ISEL
VSEL
PDOSEL
V5V
ISNP
3*2
Q6
SI2318
gate rated 20V
thresh 2.1V
4K7
R7
TVS3
15V
R8
4K7
TVS2
36V
rated 60V max
DMTH64M2LPDWQ-13
Q4B
Q4A
this protects the AP53781
from high battery voltage
SIS862ADN-T1-GE3
Q5
F1
20x5MM 7A
24V
C12
470u
C13
470u
DC-SRC
R9
4K3
R11
100Ω
R10
100Ω
3A option
24K
R13
2K
R14
USB-I-SEL
SJ9
5A option
R15
51K
select 15V
R16
1K
R17
100K
select 20V
(39K eqv)
R18
56K
R19
51K
select 28V
(26K eqv)
R20
1K
SJ8
USB-V-SEL-20V
USB-V-SEL-28V
100n
C22
JP5